Stability, Concentration, and Resonance

A Framed Dynamical Reinterpretation of
The Spike, the Sparse and the Sink: Anatomy of Massive Activations and Attention Sinks

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Abstract

This paper offers a disciplined reinterpretation of the empirical phenomena described in *The Spike, the Sparse and the Sink: Anatomy of Massive Activations and Attention Sinks*. Its purpose is not to replace the original paper's experiments, nor to inflate transformer-side observations into a universal metaphysical claim. Instead, it introduces a framed ontological–dynamical vocabulary and uses an explicit translation layer to map that vocabulary onto the original paper's observables. The central claim is deliberately restrained: the paper's reported phenomena — massive activations, sparse concentration, and attention sinks — can be read as compatible manifestations of concentration dynamics in an iterative nonlinear representational system with explicit normalization. The paper carefully distinguishes what is defined, what is observed, what is interpreted, what is only analogical, and what remains unproven.

1 Introduction

The paper *The Spike, the Sparse and the Sink: Anatomy of Massive Activations and Attention Sinks* studies three empirically reported phenomena in transformer models:

(i) unusually large activation magnitudes in a small subset of channels,

(ii) sparse or low-dimensional concentration of relevant effects,

(iii) recurrent attention bias toward specific positions, commonly called attention sinks.

The original work is empirical and mechanistic in orientation. It measures these effects, studies their geometry, and examines their dependence on architectural choices, especially normalization. The aim of the present paper is narrower and more conceptual. It asks:

> Can the observed phenomena be read within a broader dynamical vocabulary without collapsing empirical observation into conceptual overreach?

This paper answers *yes*, but only under strict methodological discipline. It therefore proceeds by separating three levels:

L1. an ontological–dynamical frame,

L2. an empirical transformer frame,

L3. an explicit translation layer between the two.

The gain of this approach is not to supersede the original paper. The gain is to make the three reported phenomena legible in a single structural language while preserving distinctions between identity, operational proxy, and analogy.

2 Scope and Claim Boundary

This paper makes one central contribution and several non-claims.

2.1 Central Contribution

The paper proposes that the original transformer's reported phenomena can be jointly interpreted as concentration signatures in an iterative nonlinear representational system with explicit normalization and repeated interaction.

2.2 Non-Claims

To prevent category mistakes, the following are explicitly *not* claimed:

N1. The original transformer paper does not prove the ontological frame proposed here.

N2. Attention sinks are not claimed to be strict attractors in the full mathematical sense.

N3. Activation magnitude is not claimed to be literal physical energy.

N4. The present reinterpretation does not replace the original empirical analysis.

N5. The structural proposition stated later is not yet a theorem.

N6. The present paper does not claim that transformer observations immediately imply a universal law of existence.

These non-claims are part of the core method of this paper, not a disclaimer added afterward.

3 Frame Separation

The paper is organized around three explicitly distinct frames.

3.1 Frame 1: Ontological–Dynamical Frame

This frame introduces abstract terms such as state, persistence, stability, concentration, and resonance. These terms are not initially identified with any transformer-specific variable.

3.2 Frame 2: Empirical Transformer Frame

This frame contains the actual observables from the original paper, including:

- activation magnitudes,
- concentration in a small number of channels or effective subspaces,
- attention weights across positions and heads,
- variation under architecture and normalization changes.

3.3 Frame 3: Translation Frame

This frame governs how concepts from the first frame may be related to observables in the second. Every relation must be marked as one of the following:

(i) **identity**, if two terms coincide exactly,

(ii) **operational proxy**, if one term is an empirical proxy for another,

(iii) **analogy**, if the relation is interpretive rather than literal.

Remark 1. *Most links in this paper are* not *identities. They are operational proxies or analogies. This is intentional and methodologically necessary.*

4 Minimal Formal Core

4.1 State and Persistence

Definition 1 (State)**.** *A system is represented by a state variable*

$$x_t \in \mathcal{X},$$

where t indexes discrete updates and $\mathcal{X}$ is the state space.

Definition 2 (Persistence)**.** *A system persists over an interval if its trajectory remains inside an admissible region*

$$x_t \in \mathcal{V} \subseteq \mathcal{X} \quad \textit{for all relevant } t.$$

The set $\mathcal{V}$ is called the viability region. Persistence is deliberately weaker than equilibrium; a system may evolve dynamically and still persist as long as it remains viable.

4.2 Stability

Definition 3 (Stability)**.** *Let*

$$x_{t+1} = F_t(x_t), \qquad F_t \in \mathcal{F}.$$

A system is stable on a region $\mathcal{V}$ if the update rule preserves viability:

$$F_t(\mathcal{V}) \subseteq \mathcal{V} \quad \textit{for all relevant } t.$$

Remark 2. *Stability is defined here geometrically rather than energetically. This avoids prematurely forcing physical semantics into the formal core. Energy-like language may later be used as an interpretation of concentration or boundedness, but it is not the first definition of stability.*

4.3 Concentration

Definition 4 (Concentration)**.** *A concentration event occurs when system activity becomes disproportionately localized in a small subset of coordinates, directions, or subspaces relative to the ambient representation space.*

This definition does not yet specify cause. It only specifies a structural signature.

4.4 Resonance Basin

Definition 5 (Resonance Basin)**.** *A resonance basin is a region, direction, or interaction pattern toward which updates recurrently align, bias, or concentrate system activity under repeated iteration.*

Remark 3. *A resonance basin is not assumed here to be a strict attractor in the full dynamical-systems sense. It is a weaker and safer cross-frame notion.*

5 Why Explicit Normalization Is Chosen

A crucial correction in the present version concerns the boundedness assumption.

An earlier version used a broad phrase such as "boundedness or scale control." That formulation was too loose because it merged at least three mathematically distinct mechanisms:

(i) explicit normalization,

(ii) implicit scale control,

(iii) architecture-level boundedness.

These are not interchangeable. They differ in operator form, mathematical role, and inferential consequences. Since the original transformer paper places normalization — specifically pre-norm behavior — at the center of the spike-to-sink bridge, the present paper adopts the narrowest and cleanest option:

> The formal core of this reinterpretation is built around *explicit normalization*, not generic boundedness.

This choice keeps the reinterpretation closer to the original paper and avoids a false appearance of generality.

6 Explicit Assumptions

Assumption 1 (Iterative Update Structure)**.** *The system evolves through repeated updates*

$$x_{t+1} = F_t(x_t),$$

where each F_t is an admissible transformation on $\mathcal{X}$.

Assumption 2 (Nonlinearity)**.** *The update family $\mathcal{F}$ contains nonlinear transformations, so the dynamics cannot be reduced to a single global linear map.*

Assumption 3 (Explicit Normalization)**.** *The update process includes an explicit normalization operator acting on intermediate representations. Schematically,*

$$\widetilde{x}_t = \mathrm{Norm}(x_t),$$

where Norm *is applied as part of the iterative computation.*

Assumption 4 (Repeated Interaction)**.** *Different coordinates, positions, or components interact repeatedly across updates.*

These assumptions are intentionally narrow. They do not attempt to cover all possible architectures. They are selected because they align with the original paper's own empirical and mechanistic emphasis.

7 Structural Proposition

Proposition 1 (Concentration Compatibility Under Iterative Nonlinear Explicitly Normalized Dynamics)**.** *Consider a system satisfying the four assumptions above. Then it is structurally compatible for the following patterns to emerge in representation space:*

(i) concentration of magnitude in a small subset of coordinates or directions,

(ii) dominance of a small number of effective subspaces,

(iii) recurrent interaction bias toward a small subset of positions or states.

Remark 4. *This is a* ***structural proposition****, not yet a theorem. It says that the three types of concentration are conceptually and formally compatible with the assumed update regime. It does* not *yet prove that every such system must exhibit these phenomena, nor that the original paper's architecture satisfies the sufficient conditions of a stronger formal theorem.*

8 What the Proposition Says and Does Not Say

8.1 What It Says

The proposition says that concentration in magnitude, concentration in subspace, and concentration in interaction need not be treated as isolated curiosities. They can be described in a common structural vocabulary once one assumes iterative nonlinear dynamics, explicit normalization, and repeated interaction.

8.2 What It Does Not Say

The proposition does not say:

- that the three observed phenomena are identical,
- that there is one unique causal mechanism behind them,
- that concentration implies strict attractor theory,
- that the present paper has proven a full formal result.

This distinction is necessary. Without it, the paper would drift from reinterpretation into unsupported mechanism-claiming.

9 Translation Layer

To prevent frame confusion, Table 1 marks each relation explicitly.

Table 1: Translation layer between the ontological–dynamical frame and the transformer frame.

Framed Term	**Transformer-Side Observable**	**Relation Type**
State	Hidden state / residual stream vector at a given layer and token position	Operational proxy
Persistence	Bounded representational evolution across layers	Operational proxy
Stability	Viability-preserving layerwise evolution under repeated updates	Operational proxy
Concentration	Large activation magnitude concentrated in few coordinates or directions	Operational proxy
Dominant directions	Sparse or low-dimensional effective subspaces	Operational proxy

Framed Term	Transformer-Side Observable	Relation Type
Resonance basin	Recurrent bias of interaction toward specific positions or tokens	Analogy
Attractor-like tendency	Stable concentration tendency in repeated interaction geometry	Analogy

Remark 5. *The most important discipline in the translation table is this:* ***resonance basin is not identical to attention sink****. The relation is analogical rather than literal. The same caution applies to any temptation to equate representational magnitude with physical energy.*

10 Restating the Original Paper in Transformer Terms

Before interpreting the original paper, it is useful to restate its three main reported phenomena strictly in the empirical transformer frame.

10.1 Massive Activations

The paper reports unusually large activation magnitudes concentrated in a small subset of channels.

10.2 Sparse Structure

The paper reports that relevant effects are concentrated in a small number of channels or low-dimensional effective subspaces rather than uniformly distributed across the entire representation.

10.3 Attention Sinks

The paper reports that some token positions receive disproportionate attention from many other positions, heads, or queries.

These three phenomena are the empirical starting point. They are not defined by the present paper; they are taken from the original one.

11 Framed Reinterpretation of the Three Phenomena

11.1 Massive Activations as Concentration Events

The safest reinterpretation is:

> Massive activations may be read as localized concentration events in representation space.

This is deliberately weaker than saying they *are* physical energy spikes. The present paper does not require that stronger claim.

11.2 Sparse Structure as Dominant Effective Directions

The safest reinterpretation is:

> Sparse structure may be read as the dominance of a small number of effective directions or effective subspaces.

This respects the original paper's geometric findings without imposing a stronger ontology than the evidence supports.

11.3 Attention Sinks as Resonance-Like Basins

The safest reinterpretation is:

> Attention sinks may be read as resonance-like basins in the repeated interaction geometry of the representational system.

This again is intentionally weaker than a full attractor claim. It retains explanatory usefulness while remaining conceptually honest.

12 Unified Reinterpretation

The three observed transformer phenomena can now be placed into a common framed vocabulary.

Observed Transformer Term	**Disciplined Framed Reinterpretation**
Massive activations	Localized concentration in representation space
Sparse structure	Dominance of a small number of effective directions or subspaces
Attention sinks	Resonance-like basins in repeated interaction geometry

The value of this reinterpretation is not that it empirically proves more than the original paper. Its value is that it renders the three effects structurally commensurable without pretending that they are identical.

13 What This Reinterpretation Explains

This paper helps explain the following:

(i) why concentration in activation magnitude, concentration in subspace, and concentration in interaction can be discussed in one structural vocabulary,

(ii) why the coexistence of these three patterns is conceptually compatible with an iterative nonlinear explicitly normalized system,

(iii) why the original paper's findings need not be dismissed as random artifacts even when some observables are representation-sensitive.

14 What This Reinterpretation Does Not Explain

The present reinterpretation does not explain:

(i) the unique causal mechanism behind each reported effect,

(ii) whether the same effects survive arbitrary basis change,

(iii) whether a strict theorem can be proven for the original architecture,

(iv) whether the same framed reinterpretation extends unchanged to all neural architectures,

(v) whether the broader ontological frame is universally valid outside the narrow translation layer used here.

These limits are not weaknesses to hide. They are conditions required to keep the paper conceptually exact.

15 Why the Present Version Is Tighter

The present version is tighter than a looser concept note for three reasons.

15.1 First, It Separates Frames

It does not move directly from transformer observables to universal ontology. It inserts a translation layer and marks each relation type explicitly.

15.2 Second, It Removes the Loose Boundedness Assumption

Instead of using a vague phrase such as "boundedness or scale control," the paper chooses the narrower and paper-faithful assumption of *explicit normalization*. This avoids mixing distinct mechanisms.

15.3 Third, It Distinguishes Proposition from Theorem

The paper does not pretend that the current structural proposition is already a theorem. It says exactly where the formal gap remains.

16 Open Formalization Path

The present paper stops at a structurally disciplined reinterpretation. A next-step technical paper would need to strengthen the formal side by doing the following:

Step 1. specify a precise class of explicitly normalized update operators,

Step 2. define a concrete scalar quantity measuring concentration or boundedness,

Step 3. prove that repeated nonlinear explicitly normalized interaction induces concentration patterns for that class,

Step 4. connect the resulting theorem back to measurable transformer quantities such as activation concentration and sink-like attention bias.

This is the correct site of future theorem-building. It is also the precise place where the present paper stops.

17 Conclusion

This paper offers a framed dynamical reinterpretation of *The Spike, the Sparse and the Sink* without confusing reinterpretation with proof. Its main result is conceptual discipline.

Within that discipline:

- massive activations are read as localized concentration events,

- sparse structure is read as dominance of effective directions or subspaces,
- attention sinks are read as resonance-like basins in repeated interaction geometry.

The strength of this formulation lies in what it refuses to do. It does not treat analogy as identity. It does not treat transformer-specific observation as universal theorem. It does not treat a structural proposition as though it were already formally proven.

What it does provide is a tighter scaffold for reading the original paper: a way to understand its three reported phenomena as related signatures of concentration dynamics in an iterative nonlinear explicitly normalized representational system, while keeping explicit track of what is defined, what is observed, what is inferred, and what remains to be proved.